JESICA MARIA RAMIREZ TOSCANO

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Applied Data Scientist with +7 years of experience in statistical data analysis, experimental design and machine learning for marketplace-quality and regulatory decisions. Focused on measuring whether products, policies, and AI systems work, especially in the public interest.

PROFESSIONAL EXPERIENCE

Uber Technologies Inc. ***September 2021 – Present***

Scientist II *Earner Progression Science*
October 2024 – Present

- **Causal measurement:** Built a causal estimation framework (order-level fixed-effects regression) that quantifies the marketplace cost of courier behavior and informs earner-quality policy.
- **Experimentation:** Designed experiments and behavioral measurement frameworks across Product, Engineering, and Operations; contributed to a real-time feedback system generating $36M in annual cost savings.
- **Multi-armed bandit:** Ran a bandit test across eight message variants to nudge couriers; lenient messages improved behavior, while punitive ones only drove churn.
- **Headroom analysis:** Used uplift modeling to target fewer earners, and analyzed changes to earner-quality metric thresholds to improve marketplace outcomes.

Applied Scientist II *Policy Research – Legal Data & Economics*
September 2021 – October 2024

- **ML and LLM classification for regulatory compliance:** Replaced manual and keyword-search review of incidents for Australian regulators (1% precision, 50% recall) with a logistic-regression text classifier (17% precision, 89% recall; projected savings of AUD 500,000 per year), then with an LLM-based (Gemini API) classifier behind a PII-redaction layer (24% precision, 91% recall). Nominated for the 2023 Reimagine Award.
- **Evidence for regulators:** Developed multi-scenario economic analyses for the New York Attorney General and statistical analyses for regulators across jurisdictions, as externally auditable evidence.
- **Self-serve tools for Legal:** Built a scenario simulator that lets Legal directors estimate litigation exposure under different assumptions, and a driver-activity dashboard that 10+ paralegals use to self-serve a driver's history and activity over any period.
- **Regulatory data productions:** Built a privacy framework for datasets produced to regulators and external parties; modeled litigation risk and financial exposure.

Banco de México (Central Bank) ***August 2017 – August 2019***
Economist *General Directorate of Economic Research*

- Built an NLP policy-tone index across 80+ central-bank statements (2008–2019) and derived inflation expectations and interest-rate forecasts (1 month to 3 years) from swaps, bonds, and surveys, supporting monetary policy decisions.

Deep Dive ***June 2020 – August 2020***
Data Scientist: helped build the company's NLP pipeline for large-scale text data.

EDUCATION

University of Chicago *September 2019 – June 2021*
Master of Science in Computational Analysis and Public Policy *With Honors*
Joint Degree by the Department of Computer Science and the Harris School of Public Policy. Coursework: Computer Science with Applications I & II, Databases, Cloud Computing, Linear Algebra, Machine Learning, Advanced Machine Learning, Data Visualization, Advanced Statistics, Causal Inference

Universidad de las Américas Puebla, Mexico *August 2012 – June 2017*
Bachelor's Degree in Economics *Magna Cum Laude*
First Place, Empirical Research Thesis Regional Competition in Puebla

SKILLS

Programming Languages: Python, SQL, R, C#, JavaScript
Data and Dev Tools: Presto, Spark, Docker
ML Tools: PyTorch, scikit-learn, transformers, LLM APIs

PROJECTS

Machine Learning

- **COVID-19 in Mexico (health):** Predicted severe outcomes (hospitalization, death) with balanced and weighted random forests on government data, informing a relative risk index across states. *Python.*
- **Predicting Polarization from News Tweets:** CNN classifying polarization in comments on news-outlet tweets. *Python, PyTorch.*

Full-Stack Development

- **TenisLAT:** Built a bilingual web app for amateur tennis leagues in Mexico City, with a Glicko-2 rating engine. *TypeScript, Next.js, Prisma, PostgreSQL.*
- **Mapping Crimes in Mexico City:** Crime-risk web app by location and time of day, from 2M+ city records and Google Maps API. *Python, Django, SQLite.*

Data Visualization

- **Chronic Crime Behavior:** Interactive, scroll-driven visualization of arrest histories across mental health and social dimensions, using the BJS 2016 Survey of Prison Inmates. *JavaScript.*

Data Pipelines and Time Series

- **Hohonu Water-Level Monitoring:** Pipeline to clean, calibrate, and predict water levels; scraped 200+ URLs for datums; improved VAR forecasts.